

L. T. MINH TRINH

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EDUCATION

Imperial College London
MSc in Artificial Intelligence

London, United Kingdom
Sep. 2026 - Present

University College London
BEng in Mechanical Engineering, Minor in Intelligent Systems

London, United Kingdom
Sep. 2023 - Jun. 2026

- Grade: First Class Honours
- Dissertation: *Continuous-Time Recurrent RL for USV Navigation (full report)*

The National Mathematics and Science College
A Level

Coventry, United Kingdom
Sep. 2021 - Jul. 2023

- Grade: A*, A*, A (in Maths, Further Maths, and Physics respectively)
- Awards: Gold (UKMT Maths Challenge), Silver (Physics Challenge), Bronze I (BPhO), Merit (BMO)

PROJECTS

Independent Research
Empirical Evaluation of Heavy-Tailed RMT Shrinkage

London, United Kingdom
Aug. 2026 - Present

- Derived the Stieltjes transform and Marchenko-Pastur spectral limits for sample covariance matrices under heavy-tailed (Student-t) returns, extending classical Ledoit-Wolf and Bouchaud RIE shrinkage to a non-Gaussian regime and benchmarking recovery via Stein's loss across concentration ratio and tail index.
- Formulated eigenvalue shrinkage as a monotonicity-constrained regression problem, training a LightGBM estimator against the derived oracle shrinkage function to identify regimes of ML over- and under-performance relative to the analytical baselines.

Dissertation
Continuous-Time Recurrent RL for Dynamic USV Navigation

London, United Kingdom
Sep. 2025 - May. 2026

- Formulated sequential decision-making under partial observability as a POMDP, training a continuous-time recurrent policy (CfC+PPO) to infer latent state from noisy signals, reducing failure rate by up to 40% relative to LSTM/GRU baselines with statistically validated results ($z = 2.57$, $p \approx 0.010$).
- Isolated performance drivers via three independent ablation studies, quantifying a bias-variance-style trade-off where curriculum training improved standard-domain robustness at a measurable cost to out-of-distribution generalisation (72.6% vs. 66.6% success rate).

Summer Research (NeurIPS Workshop - CogInterp 2025)
Metacognitive Sensitivity for Test-Time: Dynamic Model Selection

London, United Kingdom
Jun. 2025 - Aug. 2025

- Developed metacognitive framework for test-time model selection using contextual bandits (UCB/TS), improving pre-trained image classifier accuracy by 10% and vision-language models (CLIP/ALIGN) by 2% without any fine-tuning.
- Accelerated Bayesian metacognition models from CPU-bound PyMC to GPU-based NumPyro, achieving 3-5x speed-up and enabling real-time model selection.

EXPERIENCE

Autonomous and Control Engineer
UCL Racing - Mars Rover team

London, United Kingdom
Oct. 2023 - Jun. 2026

- Evaluated and iterated across SLAM approaches (VSLAM, FAST-LIO, LIO-SAM) to construct a real-time point cloud map for 3D localisation, and integrated an A*-based Nav2 path planning stack with costmap tuning for autonomous obstacle avoidance.
- Developed computer vision system for Mars rover competing in European Rover Challenge (ERC), fine-tuning YOLO for obstacle classification and implementing ArUco marker detection for autonomous navigation and collision avoidance.

TECHNICAL SKILLS

Languages

Python, C/C++, SQL, MATLAB

Frameworks & Libraries

PyTorch, JAX, Scikit-learn, NumPy/Pandas, RLlib, Gym, SciPy

Tools

Docker, Git, Linux, ROS